

Modeling of the High-Temperature Polymerization of Methyl Methacrylate. 1. Review of Existing Models for the Description of the Gel Effect

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High-temperature polymerization has recently become of interest for industry because it yields higher reaction rates, and thus less reaction time, lower viscosity, and higher conversions. However, it also leads to new challenges for process engineering in regards to materials, reactor design, and, most of all, process modeling. Because practically all kinetic research in the field of polymerizations has been realized for rather low temperature ranges (mostly below the glass transition temperature), the correct modeling of a high-temperature polymerization requires the investigation of various parameters, such as thermal initiation effects, depropagation reactions, etc., and a careful adaptation of existing models for kinetics at significantly higher temperatures. The attention is focused on the modeling of the gel effect based on existing models available from the literature. Therefore, two different types of models, one semiempirical and one based on the free-volume theory, are examined for their applicability to high-temperature polymerization by comparison with the experimental results from differential scanning calorimetry batch polymerizations. Model fitting was realized using the software package PREDICI by CiT GmbH, Rastede, Germany.

Introduction

There is a need to establish a complete model for the description of the high-temperature polymerization of methyl methacrylate. Because models are supposed to be applied to pilot- and industrial-scale polymerizations, two prerequisites need to be fulfilled. First, they should be as simple as possible to save calculation time and to be able to easily fit different experimental conditions. Second, common technical product qualities in regards to monomer and initiators rather than highly purified educts should be considered because this reflects the situation for large-scale, continuous processes. In particular, contact of the monomer with air, leading to impurities that influence the initiation behavior at high temperatures, cannot be neglected.

In this part 1 of a series of papers, the attention is focused on finding a suitable approach for the correct description of the gel effect, i.e., the change of the termination rate constants with conversion. The number of gel effect models in the literature is quite high. However, when it comes to polymerizations above the glass transition temperature T_g , the choice becomes rather poor. Additionally, most of the models have been derived from the free-volume theory by Vrentas,¹ such as the ones of Chiu et al.,² Kiparissides and co-workers,^{3,4} and Hoppe and Renken,⁵ with the aim of having a fundamental model without fitting needs but requiring the knowledge of various different physical–chemical and monomer-specific constants. These models were not taken into account for those reasons and because of their computing complexity for polymerization processes.

The choice was finally narrowed to two different model types, one semiempirical approach by Fleury⁶ and one model derived by Weickert,^{7–9} which basically uses the

free-volume theory but with fitting parameters rather than a set of specific values of the free volume. Both models allow the calculation of the termination rate change with conversion and were included in the below-mentioned kinetics scheme for high-temperature polymerization.

High-Temperature Polymerization Kinetics

Apart from the gel effect modeling, there are two major differences between polymerization kinetics below T_g and the kinetics at temperatures significantly above T_g :

(i) Initiation reactions by impurities have to be taken into account to meet the rather high initial reaction rate at high temperatures, which is considered to be caused by peroxide-containing species that form during heating cycles from dissolved molecular oxygen and decompose quickly at high reaction temperatures.^{10,11} In fact, in the absence of any initiator, methyl methacrylate (MMA) rapidly polymerizes up to high conversions at temperatures above 100 °C.

(ii) A depolymerization step becomes necessary at elevated temperatures because of the rather strong conversion limitation for MMA. The ceiling temperature for MMA is reported to be between 200 and 220 °C, i.e., quite low.

The following reaction steps are, therefore, considered in the modeling of the high-temperature MMA polymerization:

reaction step	rate constant
initiation by an initiator (DTBP)	$I \rightarrow 2R^\bullet$ $R^\bullet + M \rightarrow P_1^\bullet$ k_d, f k_p
initiation by an "impurity"	$Imp \rightarrow 2R^\bullet$ $k_{d,i}$
propagation	$P_n^\bullet + M \rightarrow P_{n+1}^\bullet$ k_p
depropagation	$P_n^\bullet \rightarrow P_{n-1}^\bullet + M$ k_{dp}
termination by disproportionation	$P_n^\bullet + P_m^\bullet \rightarrow P_n + P_m$ k_{td}
termination by combination	$P_n^\bullet + P_m^\bullet \rightarrow P_{n+m}$ k_{tc}
transfer to monomer	$P_n^\bullet + M \rightarrow P_n + P_1^\bullet$ $k_{tr,m}$

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Unfortunately, because of secrecy agreements, no further information can be provided concerning the initiation by impurities or their formation in the reactive system. Experiments have shown, however, that they affect mostly the initial reaction rate because of their fast decomposition. For the evaluation of gel effect models, it is thus of minor importance.

Models

Two fundamentally different types of models have been investigated: First of all, an approach based on the correction function of the termination rate constant

$$\frac{1}{k_t} = \frac{1}{k_{t,0}} + \frac{\gamma}{\exp(\alpha - \beta X)} \quad (1)$$

is employed for the description of the gel effect. This approach was derived from a simplified diffusion model and adapted to the MMA polymerization above T_g by Fleury.⁶ The semiempirical parameters α and β depend on the reaction parameters temperature T , initiator concentration $c_{i,0}$, and solvent content w_s (see eq 2) and determine both the onset point (α and β) and the intensity (only β) of the gel effect. Their mathematical descriptions result from various experiments that Fleury carried out for different conditions (solvent and initiator concentrations and temperature). The parameters α_0 and β_0 were originally calculated from a diffusion approach; however, in this work, they are directly adapted to experimental data (see the experimental data).

$$\begin{aligned} \alpha &= \alpha_0 - 10.9w_s + \ln([I_0]/0.002)^{0.56} \\ \beta &= \beta_0 - \frac{T - T_g}{9.3} - 38.3w_s + 2.32(1 - e^{(-[I_0]/0.002)/3.34}) \end{aligned} \quad (2)$$

This temperature dependence is logical because it has an effect on the molecular weight and viscosity. It will be shown later that also the parameters α_0 and β_0 are temperature-dependent. The initial initiator concentration $[I_0]$ is a measure of the total amount of radicals and expresses the influence of the molecular weight. Last but not least, the viscosity is also directly connected to the solvent quantity w_s .

Fenouillot et al.¹³ modified this model mathematically to

$$\frac{1}{k_t} = \frac{1}{k_{t,0}} + \frac{1}{k_{t,0} \exp[\beta(X_c - X)]}; \quad X_c = \frac{\alpha - \ln k_{t,0}}{\beta} \quad (3)$$

with $X_c = f(T, [CT])$ and $\beta = f(T)$. Furthermore, they replaced the initial initiator concentration $[I_0]$ by the chain-transfer agent concentration $[CT_0]$ with the explanation that the dependency of X_c and α on the molecular weight is governed by the chain-transfer agent rather than the initiator concentration (see eq 4).

$$\begin{aligned} X_c &= 4.289 - 0.05799T + \\ &0.00020422T^2 + 0.11 \ln(1000[CT_0] + 3) \end{aligned}$$

$$\beta = -17.85 + 0.5756T - 0.002519T^2 \quad (4)$$

Because in this work no chain-transfer agent is added, the Fenouillot model is not considered for further

evaluation. However, it should be pointed out that it might serve as an alternative to the Fleury model for chain-transfer polymerizations.

Second, a model based on the free-volume theory by Vrentas¹ was investigated. However, no "true" free-volume concept like the one of Kiparissides and Achilias^{3,4} or Hoppe and Renken⁵ was chosen because, as was already mentioned, it requires exact knowledge of various molecule-specific diffusion and free-volume coefficients. A model that combines the theoretical background of the free-volume theory with numerical parameter estimation for a better description of the experimental results is the model described by Weickert (Table 2).⁷⁻⁹ The principle of this model is the separation of the termination rate constant into a three-step process: an intrinsic diffusion step $k_{t,R}$ (arrangement of the radical chain ends to make the reaction possible), a translational diffusion k_{TD} (movement of the macroradicals toward each other), and a reaction diffusion part or "residual termination rate constant" k_{RD} :

$$k_t = \frac{1}{\frac{1}{k_{t,R}} + \frac{1}{k_{TD}^*}} + k_{RD} \quad (5)$$

The translational diffusion coefficient k_{TD}^* is molecular weight dependent and can be calculated by a "diffusion approach":

$$k_{TD}^* = \frac{k_{TD}}{M_n}; \quad k_{TD} = D \exp\left(-\frac{g_1}{V_f}\right) \quad (6)$$

Because only the apparent termination rate constant k_t can be measured, $k_{t,R}$ has to be calculated for conversions tending to zero. For eq 5, in combination with eq 6, we obtain at initial conditions

$$k_{t,0} = \frac{1}{\frac{1}{k_{t,R}} + \frac{M_{n,0}}{k_{TD,0}}} + k_{RD,0} \quad (7)$$

$$\frac{1}{k_{t,R}} = \frac{1}{k_{t,0} - k_{RD,0}} - \frac{M_{n,0}}{k_{TD,0}}; \quad k_{TD,0} = D \exp\left(-\frac{g_1}{V_{f,0}}\right) \quad (8)$$

M_0 corresponds to the initial molecular weight and is an arbitrary value that is not further specified by the authors. In this work, it is set to $M_0 = 400$ g/mol. The reaction diffusion k_{RD} is, according to Weickert, proportional to the frequency of monomer addition to the radical ends and can thus be expressed by

$$k_{RD} = Ak_p[M] \quad (9)$$

Substitution of eqs 6 and 8 into eq 5 with elimination of D leads to eq 10 (for details, see ref 9).

$$k_t = \frac{1}{\frac{1}{k_{t,0} - k_{RD,0}} + \frac{1}{g_3} [M_w^{g_4} \exp(g_1 X_{V_f}) - M_{w,0}^{g_4}]} + k_{RD} \quad (10)$$

The three parameters g_1 , $g_3 = k_{TD,0}$ and $g_4 = n$ have to be fitted to experimental data and $X_{V_f} = 1/V_f - 1/V_{f,0}$ and $V_f = \sum_i V_{f,i} \varphi_i$. Furthermore, this model takes into account the influence of diffusion limitation on the initiator efficiency using the free-volume theory by

Table 1. Kinetic Rate Constants Used in the Model

rate constant	value		ref
k_d (DTBP)	$2.8 \times 10^{14} \exp(-17617/T)$	$\text{L mol}^{-1} \text{s}^{-1}$	12
f	0.7		12
k_p	$4.9166 \times 10^5 \exp(-2191/T)$	$\text{L mol}^{-1} \text{s}^{-1}$	12
k_{dp}	$X^{eq} = 1 - \exp(23.66 - 11282.7/T)$	$\text{L mol}^{-1} \text{s}^{-1}$	
	$k_{dp} = k_p[\text{MMA}]_0(1 - X^{eq})\rho/\rho_0$		6
$k_{t,0}$	$9.8 \times 10^7 \exp(-353/T)$	$\text{L mol}^{-1} \text{s}^{-1}$	2
k_{tc}/k_{td}	$3.956 \times 10^{-4} \exp(2065/T)$		2
$k_{tr,m}$	$9.958 \times 10^5 \exp(-6197/T)$	$\text{L mol}^{-1} \text{s}^{-1}$	13

introducing a fourth adjustable parameter g_2 :

$$f = \frac{2f_0}{1 + \exp(g_2 X_{V_f})} \quad (11)$$

Experimental Part

The monomer (MMA, stabilized with 20 ppm of MEHQ, 4-Methoxyphenol) was treated with 2 M NaOH, washed several times with water, and distilled under reduced pressure to remove the stabilizer. To prevent the formation of impurities with oxygen already during storage, the monomer was stored under an inert argon atmosphere. To simulate industrial conditions, the monomer was then exposed to air prior to polymerization for saturation with oxygen. The initiator (di-*tert*-butyl peroxide, Fluka) was used without further treatment.

All polymeric samples for the evaluation of the models were polymerized in a Perkin-Elmer Pyris1 differential scanning calorimeter (DSC). The samples (approximately 20 μL) were filled in medium-pressure stainless steel crucibles with a volume of 60 μL . All experiments were conducted isothermally at the corresponding temperature, being reached within 1 min of heating.

For monomer conversion evolution, the crucibles were cooled in liquid nitrogen after given times and the sample was dissolved in 400 μL of dimethylformamide and analyzed after 1 night of stirring by purge and trap gas chromatography with butyl acetate as the internal standard.

One sample (with maximum conversion) for each temperature was dissolved in 5 mL of tetrahydrofuran (high-performance liquid chromatography grade) for molecular weight analysis and analyzed on a Viscotek TDA-300 triple-detection size exclusion chromatography.

Samples were polymerized at 140, 160, and 180 $^{\circ}\text{C}$ with an initial initiator concentration of $[\text{I}_0] = 1000 \text{ ppm}$ without additional reactants or solvents.

Modeling

All modeling was performed by using PREDICI by CiT GmbH (Germany), an industrially recognized and widely spread software package for the modeling of polymerization reactions, on a Desktop PC (Pentium 4, 2.4 GHz, 1024 MB of RAM). The advantages of using this package are its relatively easy application, numerous reactor models (from batch to continuous processes), various output possibilities, and importation of measurement data. The program also offers the possibility of carrying out multiple parameter estimations based on experimental data, guaranteeing a direct connection to the real process.

For the modeling, the above-mentioned kinetics scheme is entered, together with the corresponding rate con-

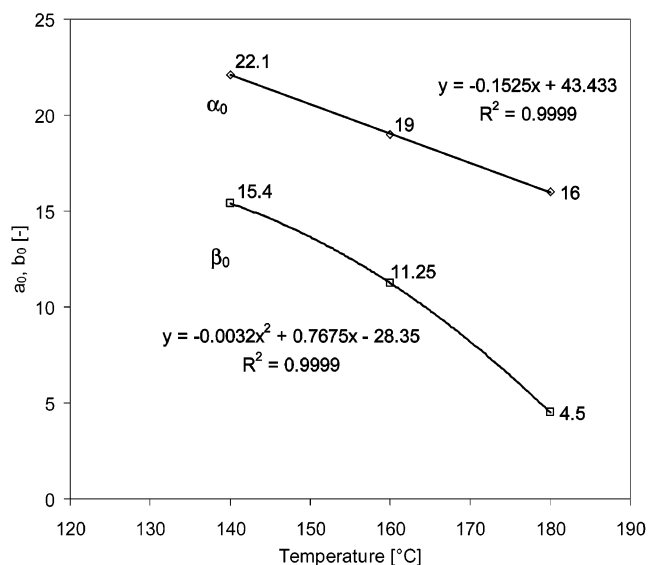


Figure 1. Temperature dependence of α_0 and β_0 for the given temperature range.

stants from Table 1 and initial concentrations, into a so-called “workshop”. For this work, a batch reactor with the volume of the DSC crucibles was defined. The gel effect models were included by separate “interpreter” files that read out the actual value for k_{tc} and k_{td} for each time step and give back the modified value corresponding to the model used (eq 3 or 10). All differential equations for the calculation of concentrations, moments, and degrees of polymerization are implemented in the software and are not visible. However, any of these values can be defined as the output per time step, together with a variety of other information. The molecular weight distribution modeling is accomplished numerically within the software by discrete Galerkin methods. The exact algorithms are protected by the creators and have not been published so far. However, there are several publications containing more information, among which the most important are refs 14 and 15.

Results

Fleury Model. For this model, the two parameters α_0 and β_0 had to be fitted to the experimental data. Although in the original model they were derived by a diffusion approach (for details, see ref 4), it was found to be more precise to fit them directly. For the three reaction temperatures, a temperature dependence was found (see Figure 1) for both parameters and determined to be

$$\alpha_0 = -0.1525T + 43.433$$

$$\beta_0 = -0.0032T^2 + 0.7675T - 28.35 \quad (12)$$

α_0 shows a linear temperature dependence, whereas β_0 exhibits quadratic behavior. Despite the model’s simplicity, the results obtained with these values show a very good prediction of the experimental results. In Figure 2, the heat flow of the polymerization at 140 $^{\circ}\text{C}$, as measured by DSC, is depicted. The model is in very good agreement with the measured points, specifically for the gel effect. As explained above, the initial reaction rate is increased by the decay of an impurity, which

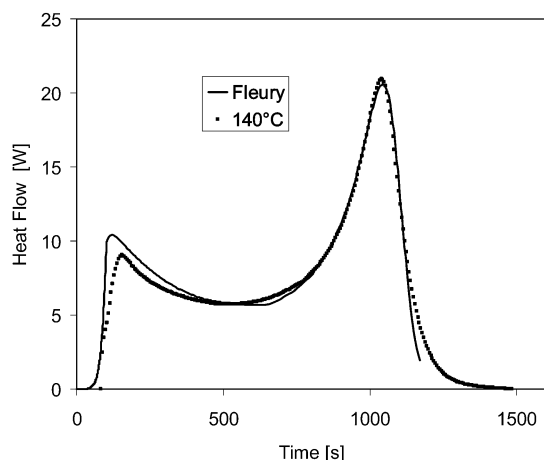


Figure 2. Comparison of the Fleury model and DSC curve for polymerization at 140 °C ($[DTBP_0] = 1000$ ppm).

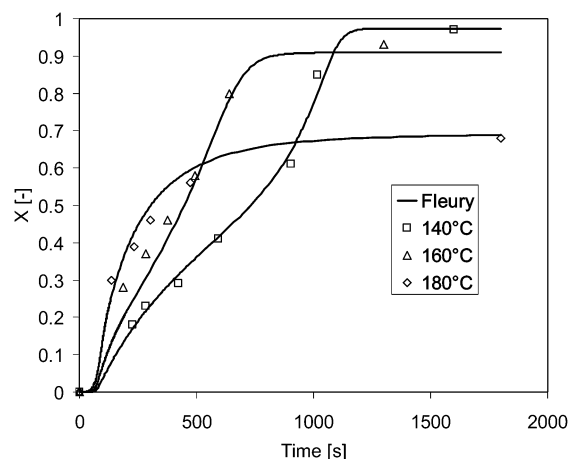


Figure 3. Monomer conversion evolution for 140, 160, and 180 °C ($[I_0] = 1000$ ppm).

adds to the radical concentration produced by the initiator and, thus, accelerates the reaction.

Figure 2 also provides a general impression of the gel effect. After the heating period, the reaction starts normally until a monomer conversion of around 40%, when the heat flow and conversion rate start to rise to a second maximum, where then almost full conversion is reached. The reason for this behavior is the reduced global termination rate with increasing viscosity due to the mobility decrease of large molecules, while the small monomer molecules keep diffusing to the macroradicals. At higher temperatures, this effect weakens because of the viscosity decrease.

Because of the exothermal character of the polymerization, there is an additional autoacceleration effect; i.e., from the point where the heat flow starts to rise, the reaction automatically goes to full conversion, which can be disastrous for reactions at larger scale. It is, therefore, essential to correctly predict the onset point of the gel effect.

Concerning the conversion evolution (see Figure 3), the model is also in good agreement with the experiments. Especially, the temperature-dependent change of the gel effect (onset and intensity) and the conversion limitation due to depropagation of the active radical chains are well-described by the model. To demonstrate the importance of the depropagation step in the kinetics scheme (see above), Figure 4 compares the conversion evolution at 180 °C with and without this step. This

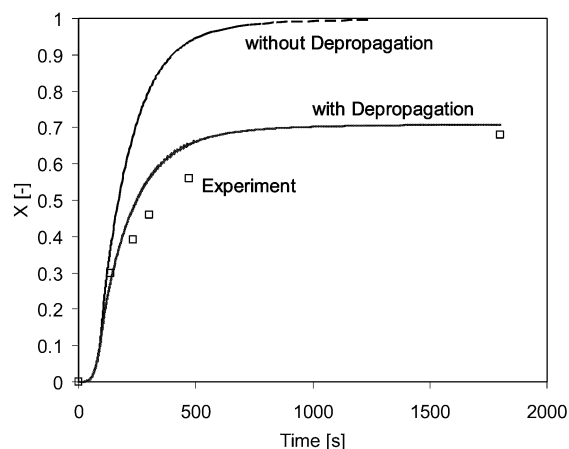


Figure 4. Monomer conversion limitation with and without a depropagation step at 180 °C ($[I_0] = 1000$ ppm).

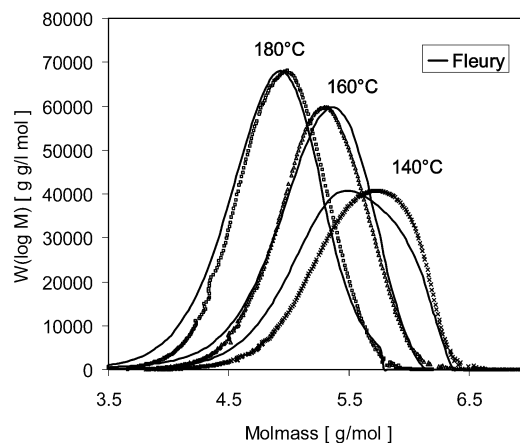


Figure 5. Molecular weight distribution comparison between experimental and predicted results.

Table 2. Parameters Needed for the Weickert Model

parameters	unit	value	fitted
$k_{RD,0}$	s^{-1}	$k_{RD,0} = Ak_p[M]_0$	
k_{RD}	s^{-1}	$k_{RD} = Ak_p[M]$	
A		1	
g_1			g_1
g_2			g_2
$k_{TD,0}$	s^{-1}		g_3
n			g_4
$M_{w,0}$	$g\ mol^{-1}$	400	
$V_{f,MMA}$	T in K	$0.025 + 10^{-3}(T - T_{g,m})$	
$V_{f,PMMA}$	T in K	$0.025 + 4.8 \cdot 10^{-4}(T - T_{g,p})$	
$T_{g,m}$	K	167	
$T_{g,p}$	K	378	
φ_i		volume fraction of i	

effect should not be neglected for the correct modeling of any MMA polymerization process at these temperatures.

An important advantage of using PREDICI is that it allows the direct prediction of molecular weight distributions. Normally, authors calculate only the mean values for M_n and M_w with the help of momentum equations. Here, we are able to determine these distributions and to compare them to the experimental gel permeation chromatography measured curves. Again, the Fleury model exhibits a good agreement with the experiment, as depicted in Figure 5, except for $T = 140$ °C, where the molecular weight is slightly underestimated.

Weickert Model. Weickert et al.⁷⁻⁹ derived their model from the theory of the free volume¹ like other

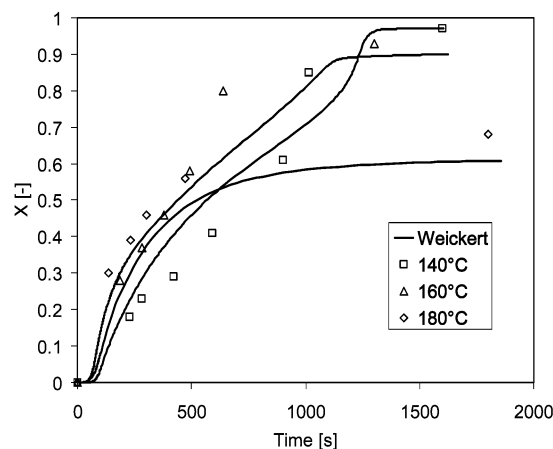


Figure 6. Monomer conversion evolution for the Weickert model compared to the experimental results.

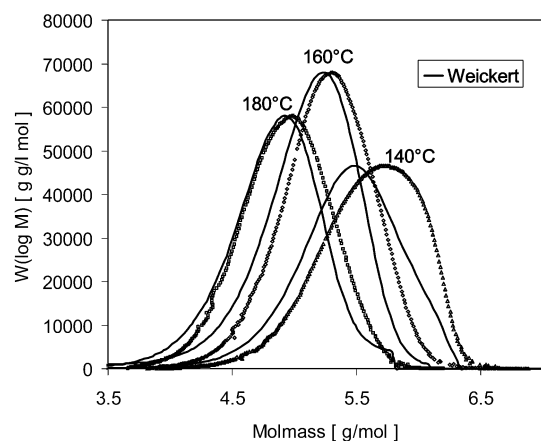


Figure 7. Molecular weight distribution predicted by the Weickert model compared to the experimental results.

Table 3. Values for Model Parameters

	original fitting [ref 9]	new fitting [this work]
g_1	1.8703 (90 °C)	5
g_2	$3.792 \exp(-746/T)$	4.5
g_3	$5.101 \exp(3211/T)$	$2.0142 \times 10^{15} \exp(-2005.3/RT)$
g_4	1	0.9957

authors before and after did. However, instead of trying to develop a model completely based on theory (where still several parameters have to be estimated), they left certain quantities open for fitting. Thus, the model can be adapted to high-temperature applications, although it was derived originally for temperatures below T_g . The results of the optimization and the original values for the four parameters g_1 , g_2 , g_3 , and g_4 are displayed in Table 3.

Still, the fitting of a model with four parameters and experimental data (conversion, molecular weight, and calorimetry) is quite complex and needs a lot of calculating power and time, especially because the simultaneous fitting of the termination rate constant and initiator efficiency is only possible with conversion and molecular weight data due to their linear dependency.

Figure 6 contains the conversion evolution graphs for the three temperatures, calculated with the newly fitted model parameters described in Table 3. The model does not fit the experimental data as well as the Fleury model does. The reaction rate in the beginning of the reaction is a little too high and the gel effect a little too late and too emphasized. Because of the strong decrease of the

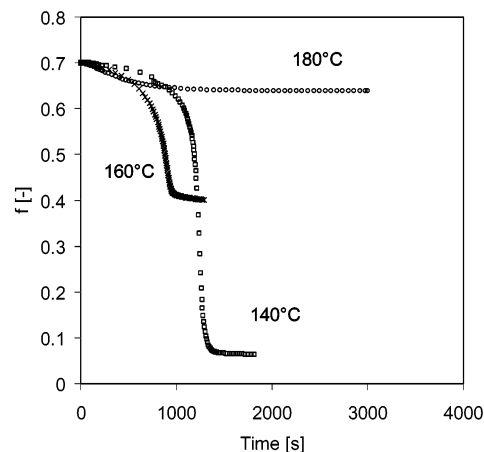


Figure 8. Change of initiator efficiency f as calculated by the Weickert model.

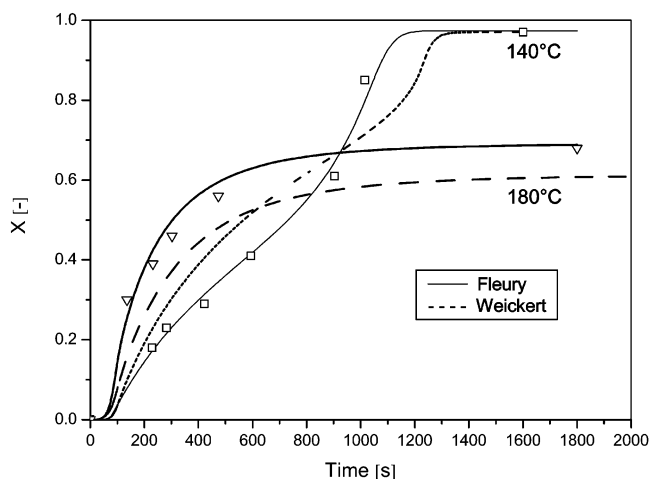


Figure 9. Comparison between the Fleury and Weickert models for monomer conversion evolution at two temperatures.

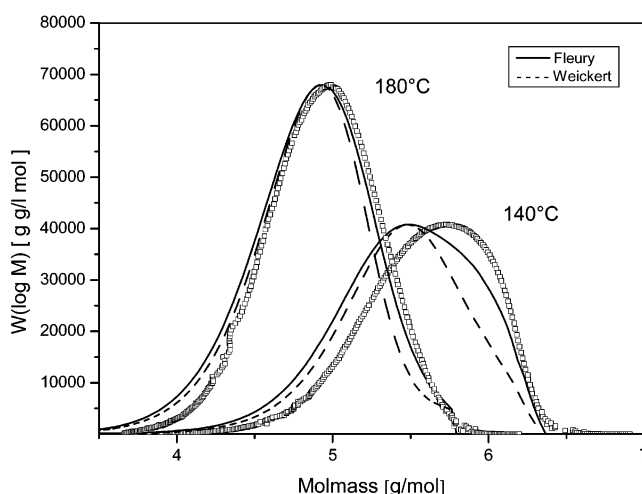


Figure 10. Comparison between the Fleury and Weickert models for molecular weight distribution.

initiator efficiency (see Figure 10), the conversion limitation is overestimated. The fitting of the molecular weight distributions depicted in Figure 7 seems to be a little more shifted to smaller molecules than those for the Fleury model.

Figure 8 shows additionally the predicted decrease of the initiator efficiency, calculated by means of the free-volume theory (see above). This decrease is another

important factor for the correct modeling of the gel effect and a clear advantage of this model. However, as the overestimated conversion limitation in Figure 6 shows, the factor g_2 will have to be further optimized.

Model Comparison. Figures 9 and 10 show a direct comparison between the two models for conversion evolution and molecular weight distributions. Although there does not seem to be a great difference between the two models for the molecular weight distribution, there are some discrepancies for the conversion evolution.

Conclusion

This review of existing models for the description of the gel effect in high-temperature polymerization was motivated by the fact that, although there are several known models available, none of them was ever tested regarding their application to an industrial- or pilot-scale process. After having sorted out models that, because of their complexity or vast number of parameters, are already not suitable, two basically different models were tested concerning their applicability to high-temperature polymerizations.

The Fleury model leads to very satisfying results and was easy to handle. The fitting of two parameters did not need too much calculating capacity for the batch modeling, which becomes an important issue for the later modeling of a continuous process. The gel effect, conversion limitation, and molecular weight distributions are precisely modeled for the investigated temperature range.

The Weickert model was adjusted to the completely different temperature range. It had been derived for below- T_g polymerizations and did not predict polymerizations above T_g correctly. After a new fitting, however, fairly good results were obtained, although there is still a certain optimization potential concerning the gel effect onset point and intensity. Because of the rather complex mathematical structure, the fitting was very time-consuming and led to many misleading results before the presented parameters were found. In any case, this model exhibits an interesting alternative to the Fleury model because it allows the prediction of the initiator efficiency decrease with conversion. Moreover, it should be mentioned that free-volume parameters are needed for this model. This could become a limiting factor for more complex polymerizations, such as, for example, copolymerizations with monomers that are not as well described in the literature as MMA.

The Fenouillot model is a direct derivation of the Fleury model, which was thought to further facilitate its handling. However, because the authors substituted the initial initiator concentration in the model parameters by the chain-transfer agent concentration, its application is limited to polymerizations with chain-transfer agents. Because one idea of the pilot-scale high-temperature polymerization is to be able to work without any additives, this model was not considered in the evaluation. It will be interesting, though, to compare the models a second time for polymerizations using chain-transfer agents.

List of Symbols

[CT] = chain-transfer agent concentration, mol L⁻¹
[I₀] = initial initiator concentration, mol L⁻¹

[MMA] = monomer concentration, mol L⁻¹

α = Fleury fitting parameter

β = Fleury/Fenouillot fitting parameter

D = diffusion coefficient, L s⁻¹

f = initiator efficiency

g_{1-4} = Weickert fitting parameters

k_d = initiator decomposition rate constant, L mol⁻¹ s⁻¹

k_{di} = impurity decomposition rate constant, L mol⁻¹ s⁻¹

k_{dp} = depolymerization rate constant, L mol⁻¹ s⁻¹

k_p = polymerization rate constant, L mol⁻¹ s⁻¹

k_{RD} = reaction diffusion limitation, s⁻¹

k_t = apparent overall termination rate constant, L mol⁻¹ s⁻¹

$k_{t,0}$ = ideal overall termination rate constant, L mol⁻¹ s⁻¹

k_{tc} = termination by a combination rate constant, L mol⁻¹ s⁻¹

k_{td} = termination by a disproportionation rate constant, L mol⁻¹ s⁻¹

k_{TD} = translational diffusion limitation, s⁻¹

$k_{t,R}$ = intrinsic termination rate constant, s⁻¹

$k_{tr,m}$ = transfer to monomer rate constant, L mol⁻¹ s⁻¹

M_n = number-average molecular weight, g mol⁻¹

M_w = weight-average molecular weight, g mol⁻¹

ρ = density, g L⁻¹

ρ_0 = initial density, g L⁻¹

T_g = glass transition temperature, K

$V_{f,i}$ = free volume of component i

w_s = solvent proportion

X^{eq} = equilibrium conversion

X_c = Fenouillot fitting parameter

Literature Cited

- (1) Vrentas, J. S. Review of the applications of free volume theory for diffusion in polymers. *Polym. Rev.* 1981, 15, 136–138.
- (2) Chiu, W. Y.; Carratt, G. M.; Soong, D. S. A computer Model for the Gel Effect in Free-Radical Polymerization. *Macromolecules* 1983, 16, 348–357.
- (3) Kiparissides, C. Polymerization reactor modeling: A review of recent developments and future directions. *Chem. Eng. Sci.* 1996, 51 (10), 1637–1659.
- (4) Kiparissides, C.; Achilias, D. Modeling of Diffusion-Controlled Free-Radical Polymerization Reactions. *J. Appl. Polym. Sci.* 1988, 35, 1303–1323.
- (5) Hoppe, S.; Renken, A. Modelling of the free radical polymerization of Methylmethacrylate up to high temperatures. *Polym. React. Eng.* 1997, 6 (1), 1–39.
- (6) Fleury, P.-A. Polymérisation du méthacrylate de méthyle à haute température. Thesis (no. 1986), Département de Chimie, École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland, 1993.
- (7) Weickert, G. Mathematical-Modeling of Polymerization Kinetics at High Monomer Conversions—a Critical-Review. *Plaste Kautsch.* 1986, 33 (8), 281–285.
- (8) Weickert, G. *Modellierung von Polymerisationsreaktoren*; Springer: Berlin, 1996.
- (9) Tefera, N.; Weickert, G.; Westerterp, K. R. Modeling of free radical polymerization up to high conversion: 2. Development of a mathematical model. *J. Appl. Polym. Sci.* 1997, 63 (12), 1663–1680.
- (10) Lehrle, R. S.; Shortland, A. A Study of the Purification of Methyl Methacrylate suggests that the “Thermal” Polymerisation of this Monomer is initiated by Adventitious Peroxides. *Eur. Polym. J.* 1987, 24 (5), 425–429.
- (11) Bamford, C. H.; Morris, P. R. The Oxidative Polymerization of Methyl Methacrylate. *Makromol. Chem.* 1965, 87, 73–89.
- (12) Brandrup, J.; Immergut, E. H. *Polymer Handbook*; John Wiley & Sons: New York, 1989.
- (13) Fenouillot, F.; Terrisse, J.; Rimlinger, T. Polymerization of methyl methacrylate at high temperature with 1-butanethiol as chain transfer agent. *J. Appl. Polym. Sci.* 1999, 72 (12), 1589–1599.

(14) Buback, M.; Busch, M.; Lammel, R. A. Modeling of molecular weight distribution in pulsed laser free-radical homopolymerizations. *Macromol. Theory Simul.* **1996**, 5 (5), 845–861.

(15) Wulkow, M. The simulation of molecular weight distributions in polyreaction kinetics by discrete Galerkin methods. *Macromol. Theory Simul.* **1996**, 5 (3), 393–416.

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